INTERNISHIP COMPLETION REPORT

Quality Control in Heavy Duty Machining equipment manufacturing

Name: Krish Bhardwaj

Class: XII C

Stream: Science

Duration: 19th May – 6th June

Company: H.C. & COMPANY

Acknowledgement

I am deeply grateful to Mr. Harish Kolambkar and my Immediate Supervisor Mr. Dheeraj Kumar Pandey, my mentors and advisors during this internship, for their invaluable advice and guidance. Their industry experience and expertise helped me to better understand the company and the industry, and allowed me to make the most of my internship.

Throughout the internship, Dheeraj Sir and Harish Sir provided me with valuable insights and guidance that helped me to navigate my tasks and responsibilities. They were always available to answer my questions and provide support, and their wisdom and expertise helped me to grow as a professional. I am thankful for their time and support, and for sharing their valuable insights with me.

I would also like to thank The Orchid School and Mrs. Anu Pazhayannur, who assisted me throughout the internship.

Information about H.C. & Company

- H.C. Company was established in 1977 as a modest tool room supplying spares to the sugar industry.
- It has transformed into a highly technical, well equipped and modernised manufacturing centre with huge capacities.
- H C & Company possesses rich industry experience in successfully handling the demands of Heavy duty as well as precision machining-based Job Works and Fabrication solutions.
- With the business operations led by experienced management the company works towards delivering end solutions in both standard as well as customized finish specifications as demanded by the customers.
- These include Shafts, Rollers, Chokes, Housing as well as other tailor-made components as required by industry sectors like Steel, Defence, Automobile, Sugar, Chemical, and others. Further, it also holds specialization in areas of producing Chokes, Sleeves, Crane Wheels, Flanges, Rings, Jigs & Fixtures as well as other tailor-made heavy-duty components.

Information about the Internship Position

- I worked under Quality Control Department as an intern.

- The working of Quality Control department includes Monitoring operations to ensure that they meet production standards.

- Recommending adjustments to the assembly or production process.

- Inspecting, testing, or measuring materials or products being produced.

- Measuring products with instruments which include, callipers, gauges, or micrometres etc.

- My role as an intern in the QC department was to observe the process of manufacturing of Heavy-Duty Machine Equipment, as per the Quality Control Plan.

- To observe, facilitate and assist during Inspection.

Work Responsibilities

- I reviewed mechanical Drawings with Dheeraj Sir.

- My responsibilities entailed –
  - Maintenance of Instrument Calibration Certificates
  - Reviewing and maintaining all documents from raw material to despatch as per ISO – 9001:2015 standards
  - Raw material physical verification-
    - Analysing defects in raw material through Non-Destructive Testing which includes **Ultrasonic Testing (UT), Magnetic Particle testing (MPT), Die Penetration Test (DPT).**

  - Observing and supervising all the parameters of the job during machining as per the drawing.
  - After machining is completed, conducting final inspection with customer.
    - Parameters for Inspection include –
      i. Reviewing reports of **Non-Destructive Testing**
      ii. **Heat Treatment**
      iii. **Machining**
      iv. **Fabrication**
      v. **Ladle Analysis**
      vi. **Hydro Test**
      vii. **Physical Testing**
      viii. **Dimensions**
      ix. **Hardness**
      x. **Smoothness etc.**

  - Inspection of Job includes-
    - Physical Verification – Analysing dimensions of the job as per drawing, (Inner Diameter (ID), Outer Diameter (OD) etc.), Surface Testing, On machine dimension parameters.

  - Instruments used in the physical inspection include-
    i. External micrometre
    ii. Internal micrometre
    iii. Bore gauge
    iv. Vernier calliper
    v. Surface Tester
    vi. Radius Gauge

- My Role included
  - Setting and Handling of the above-mentioned instruments.
  - Observing and assisting in the Job Inspection

Internship Experience and Learning

- I Understood the Working of Various types of Machines and Equipment
    - basic working of lathe machine (turning operation), Vertical Turning Lathe (VTL), Boring Machine.
    - grinding operation (Smoothness and Dimension).
    - turning operation
    - Wire cutting
- Learned The working and application of the various Mechanical Instruments.
- understood the order of various operations performed on a job strategically. (How work is planned, and tasks are organised to achieve maximum efficiency.)
- Understood and Observed Various operations such as Turning, Grinding, Stress relief, welding etc.)
- During my inspections, I interacted with extremely qualified and experienced persons in the field of manufacturing. **These gave me an insight of the internal working of the industry.**
- The soft skills I have acquired throughout the period of my internship are-
    - Effective Communication Skills
    - Interpersonal skills
    - Time Management
    - Understanding workplace culture
    - Decision making
    - Punctuality
    - Quality of work

Inspections

- Metso: Outotec

<u>Shaft Weldments</u>

Inspection & Test Plan

Item descriprtion - Weldment Shaft				Supplier	H. C. & COMPANY		Client - Metso:Outotec				PO No: 4503484517 & 4503484383
				Address	S-146, MIDC BHOSARI, PUNE 411026		ITP No.: Weldment Shaft/IR/01				PO Date: 01.09.2022
ITEM/OPERATION	OPERATION	PARAMETERS	METHOD	EXTENT OF CHECK	REFERENCE	ACCEPTANCE	RECORD	INSPECTION			REMARKS
					STANDARD/DOC	STANDARD	FORMAT	Supplier	HC	METSO	
PIPE	Raw Material Inspection	Ladle Analysis	Spectrometer	Every Heat Basis	ASTM A 106 GR B (2019)		Mill TC	P	R	R	Mill TC will be provided along with Final TC
	Raw materail Inspection & testing	Hydrostatic Test		100%	EN 10204 : 3.1	PSI 2800	Mill TC	P	R	R	
		1 Physical Testing	Tensile Test	Integral Test Piece On Per Heat per Heat Treatment basis		At Room Temperature UTS : 415 MIN YS : 240 MIN EL : 22	Mill TC	P	R	R	
	Inprocess Inspection (machining of pipe)	Dimension	Measurement	100%	Drawing	Drawing / Spec.	Diemnsion report	P	R	R	
SHAFT	Raw Material Inspection	Ladle Analysis	Spectrometer	Every Heat Basis	AISI 1018/1020/ASTM A 105/EN8		Mill TC	P	R	R	
	Forging	Temperature, Time, Reduction Ratio etc	Open Die Forging	100%	As per internal Procedures of the forger Reduction ratio shall be Min 4:1		Log Book	P	—	—	Actual Reduction Ratio will be certified in Final TC
	Heat Treatment	Normalising		100%			HT Chart	P	R	R	
	Raw materials Inspection & testing	1 Physical Testing	Tensile Test	Integral Test Piece On Per Heat per Heat Treatment basis			Mill TC	P	W/H	R	
		2 Hardness		100%					W/H	R	
		3. UT Examination (Proof Mcnd)	Pulse Echo Technique, Contact Method & Manual Scanning	100%	ASTM A388	6 mm FBH on DGS Scale	Mill TC	P	W	W	
	Inprocess Inspection (machining of	3 Dimension	Measurement	100%	As per drawing		Dimension report	P	P	R	
		4 MPI		100%	As Per Standard		MPI report	P	R	R	
	Shrink Fitting	Interference	Measurement	100%	As per drawing		DIR	.....	P	R	
Assembly of Pipe and Trunnion	Fabrication	Procedure qualification and welder's qualification	GTAW/FCAW	100%	AWS D1.1/ASME SECTION IX	APROOVED WPS, PQR, WPQ	WPS , PQR , WPQ record	.....	P	R	Die Penetrant Test 100%, Visual 100%
		Visual inspection on Weld surface	Visual	100%	AWS D1.1	AWS D1.1	Visual inspection report	P	R	R	Balance
		DP on weld surafce	DP	100%	ASME Sec V	ASME Sec VIII	DP report	P	R	R	
	Stress Relieving		GAS FIRED FURNACE	100%	14.60.00.01		Graph & TC	P	R	R	
		Visual		100%				.....	P	W	
		DPT		100%				.....	P	W	(on the weld joint between tube and trunnion)
	Final Machining	Visual/Dimensional	Apperaence / Measurement all critical dimension		As Per Drawing	As per drawing / Specification	DIR	.....	P	W	Are instruments will have valid calibration. Runout will be checked on Machine
R- Review	H-Hold										
W- Witnesss											
P- Perform							For Metso:Outotec				

H.C. COMPANY Date. Sign... Key H.C. Department, PUNE - 26

29.09.2022

TATA
TATA PROJECTS

INSPECTION REPORT

Inspection Report Number	272359-060623-01	Rev:	00	Date:	06.06.2023
Project code	408972				

I. General

Client:	Metso:Outotec	Project:		Jindal Steel Odisha Ltd. PP2, Angul	
Supplier:	H.C.& Company	Location		MIDC Pune	
Sub- Supplier:	-----	Location---------		----	
P.O. Number:	4503484517	Rev.	00	Date	25.08.2022
Sub- Supplier P.O. Number:	-----	Rev.	-----	Date	-------

II. Visit Details:

NOI/E-mail Ref:	RFI-04-02.06.2023 - Mail from Ashesh Dutta		Date:	02.06.2023
Inspection Location:	H.C. & Company S-Block Plot No 146 MIDC Pune – 411026			
Contact Details				
Supplier	Name	Mr. Dheeraj	Phone	9523408264
Sub Supplier	Name	---------	Phone	----------
Visit Date(s)	06.06.2023 (01 Day)			
Inspection Stage	Stage	Final	Pre Dispatch	
P.O. Order Status	Completed	Partially Completed		

III. a). Inspected Items (Annexure may be prepared, if required)

Sr. No. / PO Item No/BBU No	Description	P.O. Qty.	Offered Qty	Inspection Sample Qty.	Accepted Qty.	Rejected Qty.	Hold Qty.
00060	Shaft Weldment Material ZX11166990 Drg No.: 9-84310 Rev-F Sr No.: HC-40 to 46, HC-51 to 53, HC-56, HC-57, HC-60, HC-63 & HC64	72	15	15	15	00	00

b). Offered condition: Shaft weldment offered in final machine and welded condition

IV. Summary of Inspection Activities Performed:

a) Reference Documents

Document No.	Rev	Title	Approval Status	Document Supplied by
4503484517	00	PO	-----	Client
Weldment Shaft/IR/01	R1	Inspection & Test Plan	Approved	Supplier
9-84310 sheet-1	F	Drawing	Approved	Supplier

TPI 937

Format No: 15.03.01/F-01/R5 Date: 08-05-2021

M/s. H.C. COMPANY PUNE M.I.D.C., Bhosari, PUNE

TATA TATA PROJECTS	INSPECTION REPORT

14.60.00.01	-	Stress relief Specification	Approved	Supplier

b) Previous report reference, if any: IR NO 259670-250323-01 & 267417-06052023 & 260468-260523-01.

c) Inspection, Measuring and Test Equipment:

Sr. No.	Description	ID No	Make	Type	Range	Calibration Validity	Traceability to NABL/NPL
1	Measuring Tape	----	Freemans	Analog	0 – 15 mtrs	15.10.2023	Yes
2	Vernier Caliper	0113422	Mitutoyo	Digital	0 – 300 mm	15.10.2023	Yes
3	Plunger Dial Gauge	ZRD 280	Mitutoyo	Analog	0 - 10 mm	25.08.2023	Yes
4	Ext Micrometer	16075753	Mitutoyo	Analog	150 -300 mm	21.07.2023	Yes
5	Ext Micrometer	701008	Mitutoyo	Analog	300 – 400 mm	14.02.2024	Yes
7	Surface Roughness Tester	513662211	Mitutoyo	Digital	--	14.02.2024	Yes
9	Ext Micrometer	66395355	Mitutoyo	Analog	100 – 200 mm	14.02.2024	Yes
11	Bore gauge	MVCL/2022- 09/08538	Mitutoyo	Analog	50-100 mm	01.12.2023	Yes

a) Accuracy of measurements ensured - Yes ☒
b). Test witnessed at NABL Approved Laboratory - Yes ☐ No☐ NA
If Yes, Verify and note the NABL certificate number and its validity.
If No, Verify the facilities as per check list: TQS-IS-CL-02**
****This is applicable for tests witnessed at unapproved Laboratories/manufacturing facility.**
(Tick applicable clause)

d) ITP/QAP/MQP –Activities: (H-Hold, W- witness, R- Review, S- Surveillance)

ITP/QAP Clause	Inspection activity as per QAP/ITP.	H/W/ R/S	Quantum of Check
01	Pipe Raw material Inspection	R	Every Heat Basis
	Pipe Inspection and testing Hydro testing	R	100%
	Pipe Inspection and testing Physical testing	R	One per heat Basis
	Pipe inspection and testing (Dimension)	R	100%
02	Shaft Raw material Inspection	R	Every Heat basis
	Shaft Forging	R	100%
	Shaft Heat Treatment	R	100%
	Shaft Inspection and testing 1 physical testing	R	Integral test piece one/ heat treatment basis
	Shaft Inspection and testing 2 Hardness	R	100%
	Shaft Inspection and testing 3 UT Examination Proof Machined	W	100%
	Shaft 4) Dimension	R	100%
	Shaft 5 MPI	R	100%
03	Assembly of Pipe and Trunnion Shrink Fitting,	R	100%
	Assembly of Pipe and Trunnion Fabrication WPS/PQR/WPQ	R	100%
	Assembly of Pipe and Trunnion Stress Relieving – VISUAL	W	100%
	Assembly of Pipe and Trunnion Stress Relieving –DPT	W	100%
	Assembly of Pipe and Trunnion Final Machining	W	100%

TPL 937

Format No: 15.03.01/F-01/R5 Date: 08-05-2021

H.O. H.C. COMPANY PUNE Bhosari, PUNE

e) Inspection Activity Narrative Summation:

Supplier offered below items unpainted condition for inspection: 15 Nos Shaft Weldments in unpainted and Final machined condition (At J-69, MIDC Bhosari, Pune)

QAP Cl. No: -1: Pipe Raw Material Inspection, Hydro testing, Physical testing & Dimension

- MTC Reviewed as per EN10204-3.1 for mechanical test, chemical analysis, Hardness hydro & dimensions as per material specification SA106 Gr B and found satisfactory.

QAP Cl, no:2 -Shaft Raw Material Inspection (AISI 1020) Forging, Heat Treatment

- MTC / Lab test report for material as per specification AISI 1020 reviewed and found in order
- Forging, Heat Treatment report reviewed as per ITP and found in order.

QAP Cl, no: 2- Shaft Inspection and testing, Physical Testing, Hardness, UT Examination dimension & MPI

- Shaft Physical Testing for tensile test, Hardness report reviewed and found in order
- UT report for shaft in proof machined condition reviewed and found satisfactory. Vendor has not offered for UT witness & same have been discussed with Metso & advised to review vendors internal UT report. - found in order
- Shaft – Magnetic Particle testing report for machining part: Report reviewed and found satisfactory.
- Supplier Inspection report for Dimension inspection of shaft Reviewed and found satisfactory

QAP Cl No: -3 Assembly of Pipe and Trunnion-Shrink Fitting

- Assembly of Pipe and Trunnion-Shrink Fitting Report reviewed and found in order

QAP Cl No: -3 Assembly of Pipe and Trunnion-Fabrication

- WPS/PQR/ WPQ Review activity completed with previous dated 31.10.2021
- Visual: 15 Nos shaft Weldments visually checked in finish machined after Stress relieved condition and overall workmanship found satisfactory
- Reviewed DP Test of weld joint between pipe and trunnion Shaft Weldments segment and found no significant indication- satisfactory

QAP Cl No: -3 Assembly of Pipe and Trunnion-Stress Relieving

- Stress relieving – Stress Relieving Report / Chart: reviewed and found satisfactory.
- Dimensional inspection carried out (After Fabrication and Stress Relieving) of 15 Nos Shaft Weldments checked as per drawing and dimensions are found within tolerance. (witness report)
- Witnessed DP Test on weld joint between tube and trunnion Shaft Weldments segment and found no significant indication- satisfactory.

QAP Cl No: -3 Assembly of Pipe and Trunnion-Final Machining

- Shaft weldments in final machined and after stress relieved condition for final inspection dimension (15nos) and random witness 1 No Shaft weldment (9-84310 Sr No HC-41) for run out check on machine area and found satisfactory.
- 100% Shaft machine area surface roughness witnessed as per drawing and found within acceptable limit.
- 100% Total length of assembly found within found +-3 mm as per drawing.
- 100% H section dimension found within limit as per fabrication tolerance mentioned in Drg.
- Key bar used 16 mm size against 15 mm –Clearance received from Client-Metso through email.
- One no weldment shaft run out checked on machine & observed less than 0.25mm hence found Satisfactory

Remarks

- Machined surface to be protected with rust preventive coating as per specification 14.56.00.02.
- Exterior surface shall be painted with high temperature paint unless otherwise specified as per PO.
- Pack consignment suitably to prevent damage

TPL 937

H.C. COMPANY PUNE M.I.D.C. Bhosari, PUNE

Shaft Weldment Material offered for Final Dimension and Visual Inspection check, DP test & Run out test witness.

use of external micrometre to check dimensions.

Internal Micrometre and Vernier Calliper

Surface Tester and DPT

❖ L&T Defence

Breech Section

LARSEN & TOUBRO L&T Defence QUALITY CONTROL	INSPECTION CLEARANCE SLIP	ICS NO.	06L030351/ HCC001/BS/02
		DATE	06.06.2023
		PAGE NO.	1 to 21

Project No :	S067596B	Project Name :	ITTL		
Vendor Name:	H.C & COMPANY	Vendor PO No.:	06L030351/SATYAJIT PATIL,	Amendment No. & Date:	0 & DT.03.09.2021
ICLS Call No.:	06L030351-2	Approved QCP No.:	QAP-249-S067596A-B R0		

INSPECTION STAGES / DOCUMENT INDEX

SN	Inspection stages/ Documents index	Remark	Page No.
1	Raw Material Clearance: MIR, MTC,TC & MCR	NA	—
2	COC/TC Of Bought outs/FIM documents	NA	—
3	Heat Treatment Details: RM & SR report	NA	—
4	WPS/PQR/WPQ reports	NA	—
5	Part level set-up and dimension report	NA	—
6	NDT of Raw Material / Welding: DP/MT/RT/UT	NA	—
7	Fabrication VIDI of Assly/Sub-Assly/VPart/Unit/Enclosure/ Panel	NA	—
8	Machining VIDI of Assly/Sub-Assly/VPart/Unit/Enclosure/ Panel	Report No.06L030351/BSA /V06 Dt. 19.05.2023 &06L030351/BSA V06/A & B Dt. 31.05.2023	2 to 17
9	Load test reports	NA	—
10	Coating / Painting reports	NA	—
11	Visual Inspection of Surface/Unit/Enclosure/Panel	Reviewed and found in order.	–
12	Assembly Inspection / Functional Inspection	NA	—
13	Customer's/TPI's MOM & Clearance faxes	CNAI MOM Dtd. 19.05.2023 & 31.05.2023.	18-20
14	Applicable NCRs / DCRs	NA	—
15	Other documents	PHOTOGRAPH	21
16	Packing Inspection Report	Yes/No	—
17	Surface protection of machined surface with rust preventive chemical VPCI/ Black Japan	~~Yes/No~~/NA	—
All related documents Attached. **YES/NO:**		YES	**Total Pages:** 21

Customer QC / TPI Clearance (tick). **Obtained/ Not-Obtained**

Attachments of photograph of items having identification number details : **Yes / No**

ITEM DETAIL

PO Line No	PO Line Qty	Insp.Qty	Prev. Cleared Qty	Bal. Qty	Part Description	Part No.	Drawing No.	Rev	Identification
1	4	1	1	2	Breech Section	DE-4631-0	DE-4631-0	R08	Hard Punch as 06L030300 BS V06

SPECIFIC REMARK

1. Receipt inspection shall be done by concern L&T QC.
2. Balance work as per NCRs/Po/Emails.
3. Refer previous ICS No.:06L030351/ HCC001/BS/VK/01 Dtd.:25.04.2023

RELEASED FOR NEXT OPERATION: - YES / ~~NO~~

06-06-2023

AMOL THOKAL
L&T - QC

Security: Restricted Document

L&T-DEFENCE-QC-ICS-001 R02 Dt. 07.12.2021

MINUTES OF INSPECTION

Date : 31 May 23

1. **Call Ref No.** : DEF/WES/ITTL/ 1135.6/2SS/QC/SANN/12749/259 dt 27 May 23
2. **PO No** : 108592 dated 24.03.2021 (FO/1135.6)
3. **Store Inspected** : Breech Section Tube (**ID No. 06L030300 BS V06)**
4. **Qty Inspection** : 01 number.
5. **Inspection Stage** : On & Off machine dimensional inspection **(2ND Stage)**
   Drg No: DE-4631-0, R08 (Sh. 1 of 4 & Sh. 2 of 4)
6. **QAP No** : QAP-249-S067596A-B Rev 0.
7. **Qty offered** : 01 No.
8. **Place of Inspection:** M/s HC & Company, Bhosari, Pune - 411026.
9. **Date of Inspection :** 31 May 2023.
10. **DETAILS OF INSPECTION:** With reference to call letter & PO. On & Off machined visual and dimensional inspection (2ND stage) of Breech Section assembly was carried out. Details are as follows:

(a) **Visual Inspection:** Visual inspection of offered store carried out and found satisfactory. Identification No. & NAI hard stamp was observed on the stiffener. Few sharp edges, Burr were observed and found satisfactory post rework.

(b) **Dimension Inspection**: 2nd Stage On & Off machine dimensional inspection of Breech Section Assembly carried out and found satisfactory as per Rev 8, However R8 revision submitted to CNAI(W) for review & approval. (Reports enclosed)

(c) Fitments of threaded part checked with Locking Ring (ID No. DE-4634-0-1 KI-6) for dimensions of threaded part fitment checked with Breech section assembly and found satisfactory.(Ref. CNAI clearance letter No. BI/2759/6 Dated 04 Feb 23).

(d) Few machining operations like Tapping, Helicoil inserts, match-drilling were not carried out; same to be offered for inspection during final assembly/integration.

(e) Identification No. **06L030300 BS V06** hard Punched on outer surface of the tube and observed with NAI hard stamp. (Photographs enclosed)

(f) **Document Review.**
(i) Document review of raw material carried out vide Mom dated 15 Jan 23.
(ii) Clearance letter for forgings, Letter no. BI/2759/6 dated 06 Sep 22.
(iii) Previous stage MoM dated 21 Mar 23 (for DPT, DI & hydro test).
(iv) 1st stage on & off machine visual and dimensional inspection carried out on 19 May 23.

31.05.23	31.05.2023	31.5.23
(Dheeraj Pandey)	(Amol Thokal)	(Mahesh Ghule)
QC-Engineer	QC-Engineer	Rep. CNAI(W)
for M/s HC & Co.	for M/s L&T-Defence	

H.C. COMPANY PUNE Bhosari, PUNE-26

Middle Section

LARSEN & TOUBRO L&T Defence QUALITY CONTROL	INSPECTION CLEARANCE SLIP	ICS NO.	06L042793/HC001/MS/06
		DATE	04.06.2023
		PAGE NO.	1 to 13

Project No :	S067596B	Project Name :	ITTL		
Vendor Name:	H.C & COMPANY	Vendor PO No.:	06L042793/SATYAJIT PATIL	Amendment No. & Date:	DT. 02.06.2023
ICLS Call No.:	06L042793-2	Approved QCP No.:	QAP-249-S067596A-B R0		

INSPECTION STAGES / DOCUMENT INDEX

SN	Inspection stages/ Documents index	Remark	Page No.
1	Raw Material Clearance: MIR, MTC,TC & MCR	NA	—
2	COC/TC Of Bought outs/FIM documents	NA	—
3	Heat Treatment Details: RM & SR report	NA	—
4	WPS/PQR/WPQ reports	NA	—
5	Part level set-up and dimension report	NA	—
6	NDT of Raw Material / Welding: DP/MT/RT/UT	NA	—
7	Fabrication VIDI of Assly/Sub-Assly/VPart/Unit/Enclosure/ Panel	NA	—
8	Machining VIDI of Assly/Sub-Assly/VPart/Unit/Enclosure/ Panel	Report No.06L042793/MS/02 Dt. 08.05.2023	2 to 10
9	Load test reports	NA	—
10	Coating / Painting reports	NA	—
11	Visual Inspection of Surface/Unit/Enclosure/Panel	Reviewed and found in order.	–
12	Assembly Inspection / Functional Inspection	NA	—
13	Customer's/TPI's MOM & Clearance faxes	CNAI MOM Dtd. 08.05.2023	11-12
14	Applicable NCRs / DCRs	NA	—
15	Other documents	PHOTOGRAPH	13
16	Packing Inspection Report	Yes/No	—
17	Surface protection of machined surface with rust preventive chemical VPCI/ Black Japan	Yes/No/NA	—
All related documents Attached. **YES/NO:**		YES	**Total Pages:** 13

Customer QC / TPI Clearance (tick): **Obtained/ Not-Obtained**	Attachments of photograph of items having identification number details : **Yes / No**

ITEM DETAIL

PO Line No	PO Line Qty	Insp.Qty	Prev. Cleared Qty	Bal. Qty	Part Description	Part No.	Drawing No.	Rev	Identification
1	2	1	1	0	Middle Section	DE-4632-0	DE-4632-0	R08	Hard Punch as 06L030300 MS V06

SPECIFIC REMARK

1. Receipt inspection shall be done by concern L&T QC.
2. Balance work as per NCRs/Po/Emails.
3. Refer previous ICS No. :06L030300/VINA002/MS/06 Dtd.02.04.2023 & 06L042793/HC001/MS/05 Dtd.01.04.2023

RELEASED FOR NEXT OPERATION: - YES / NO

04.06.2023

AMOL THOKAL
L&T - QC

Security: Restricted Document

L&T-DEFENCE-QC-ICS-001 R02 Dt. 07.12.2021

INSPECTION REPORT

Date 08 May 23

1. **Call Ref No.** :DEF/WES/ITTL/1135.6/2SS/QC/SANN/12473/243 Dated 06 May 23
2. **PO No** :108592 Dated: 24.03.2021 (P1135.6)
3. **Store Inspected** : Middle Section Assembly (ID No. 06L030300 MS V06)
4. **Qty Inspection** : 01 no. Assembly
5. **Inspection Stage** : On & Off machine dimensional inspection (final machining)
6. **Inspection Date** :- 08 May 23
7. **Drg No** : DE-4632-0, R08 (Sh. 1 of 4 & Sh. 2 of 4)
8. **QAP No** : QAP No.-QAP-249-S067596A-B, Rev 0
9. **Place of Inspection:** M/s HC & Company, Bhosari, Pune - 411026.
10. **DETAILS OF INSPECTION:**

With reference to call letter & PO, Firm offered 01 no. Middle Section Assembly for Visual, dimensional inspection (after machining) & Documents review.

Details of activities are as follows:-

(a) **Visual Inspection:** Visual inspection of offered store carried out, Assembly found with unique ID no. as ID No. **06L030300 MS V06.** Burrs were observed on machined edges, rechecked after rework and found satisfactory. Offered Middle Section Assembly found with CNAI (W) hard stamp as " " near to ID no. (Photo enclosed)

(b) **Documents Review**:-

(i) Raw material consumption plan MOM dated 17 Jan 23 review and found satisfactory.

(ii) Hydro test and DP test of the store was carried out vide inspection report dated 01 April 23 and found satisfactory.

(c) **Dimension Inspection**: On-machine and off- machine dimensional inspection of Middle Section tube was carried out and found satisfactory. (**report enclosed**).

However some dimensions are to be carried out on assemblies as mentioned in the dimension report. Same to be offered to CNAI (W) for dimensional inspection post assemblies with rest of the tubes.

Note :-Assembly will be proceed to next stage only after CNAI(W) clearance letter.

08/05/2023	08-05-2023	08 May 23
Dheeraj Pandey	Amol Thokal	(Naveen Chandara Rawat)
QC-Engineer	QC-Engineer	Chargeman(Mech)
For HC & COMPANY	For L&T	CNAI (W)

H.C. & COMPANY Date. Sign... M.I.D.C. Bhosari, PUNE - 26

Muzzle Section

LARSEN & TOUBRO L&T Defence QUALITY CONTROL	INSPECTION CLEARANCE SLIP	ICS NO.	06L042793/ HCC001/MZS/02
		DATE	02.06.2023
		PAGE NO.	1 to 10

Project No :	S067596B	Project Name :	ITTL		
Vendor Name:	H.C & COMPANY	Vendor PO No.:	06L042793/SATYAJIT PATIL,	Amendment No. & Date:	0 & DT.20.03.2023
ICLS Call No :	06L0342793-2	Approved QCP No.:	QAP-249-S067596A-B R0		

INSPECTION STAGES / DOCUMENT INDEX

SN	Inspection stages/ Documents index	Remark	Page No.
1	Raw Material Clearance: MIR, MTC,TC & MCR	NA	—
2	COC/TC Of Bought outs/FIM documents	NA	—
3	Heat Treatment Details: RM & SR report	NA	—
4	WPS/PQR/WPQ reports	NA	—
5	Part level set-up and dimension report	NA	—
6	NDT of Raw Material / Welding: DP/MT/RT/UT	NA	—
7	Fabrication VIDI of Assly/Sub-Assly/VPart/Unit/Enclosure/ Panel	NA	—
8	Machining VIDI of Assly/Sub-Assly/VPart/Unit/Enclosure/ Panel	Report No.06L042793/MZS /02 Dt. 24.05.2023	2 to 7
9	Load test reports	NA	—
10	Coating / Painting reports	NA	—
11	Visual Inspection of Surface/Unit/Enclosure/Panel	Reviewed and found in order.	–
12	Assembly Inspection / Functional Inspection	NA	—
13	Customer's/TPI's MOM & Clearance faxes	CNAI MOM Dtd. 24.05.2023	8-9
14	Applicable NCRs / DCRs	NA	—
15	Other documents	PHOTOGRAPH	10
16	Packing Inspection Report	Yes/No	—
17	Surface protection of machined surface with rust preventive chemical VPCI/ Black Japan	~~Yes/No~~/NA	—
All related documents Attached. **YES/NO:**		YES	**Total Pages:** 10

Customer QC / TPI Clearance (tick): **Obtained/ Not-Obtained**

Attachments of photograph of items having identification number details : **Yes / ~~No~~**

ITEM DETAIL

PO Line No	PO Line Qty	Insp.Qty	Prev. Cleared Qty	Bal. Qty	Part Description	Part No.	Drawing No.	Rev	Identification
2	2	1	1	0	Muzzle Section	DE-4633-0	DE-4633-0	R07	Hard Punch as 06L030300 MZS V06

SPECIFIC REMARK

1. Receipt inspection shall be done by concern L&T QC.
2. Balance work as per NCRs/Po/Emails:
3. Refer previous ICS No..06L042793/ HCC001/MZS/VK/01 Dtd.25.04.2023

RELEASED FOR NEXT OPERATION: - YES / ~~NO~~

02.06.23

AMOLTHOKAL
L&T - QC

Security: Restricted Document

L&T-DEFENCE-QC-ICS-001 R02 Dt. 07.12.2021

INSPECTION REPORT

Date: 24 May 23

1. **Call Ref No.** : DEF/WES/ITTL/1135.6/2SS/QC/SANN/12641/256 Dtd.23 May23.
2. **PO No** :108592 Dated: 24.03.2021 (P1135.6)
3. **Store Inspected** : Muzzle Section Assembly Serial No. 06L030300 MZS VO6
4. **Qty Inspection** : 1 No.
5. **Inspection Stage** : On & Off machine dimensional inspection (final machining) of Muzzle section
6. **Drg No** : DE-4633-0 R07 (Sh. 1 of 4 & Sh. 2 of 4)
7. **QAP No** : QAP No.-QAP-249-S067596A-B Rev. R0
8. **Qty offered** : 1 No.
9. **Place of Inspection:** M/s HC & Company, Bhosari, Pune - 411026.
10. **Date of Inspection :** 23 May 2023
11. **DETAILS OF INSPECTION:**

With reference to call letter, PO. and Inspection report dated 24 May 23, Visual & dimensional inspection (final stage) of Muzzle Section assembly was carried out. Details of activities are as follows:

(a) **Visual Inspection:** Visual inspection of offered store carried out, and found satisfactory. Hydro test of the store was cleared vide Dtd.21.04.2023. Identification **06L030300 MZS VO6 and NAI stamp** found hard punched on stiffener of the tube section.

(b) **Dimension Inspection**: On-machine and off- machine dimensional inspection of Muzzle Section tube was carried out and found to be satisfactory. (**Report no. 06L042793/MZS/02 dated 24 may 23 enclosed**). However some dimensions are to be carried out on assemblies as mentioned in the dimension report. Same to be offered to CNAI(W) for dimensional inspection post assemblies with rest of the tubes.

24/05/23	24.05.23	24.05.23
Mr.Dheeraj Pandey	Mr.Amol Thokal	Mr.Deepak P
QC-Engineer	QC-Engineer	ChargeMan(Mech)
For HC & COMPANY	**For L&T**	**CNAI (W)**

H.C. & COMPANY
Date.
Sign...
M.I.D.C. Bhosari, PUNE - 26

Conclusion

My first exposure to professional life through internship has benefited me mostly in strengthening my personal skills. I have developed a positive attitude, and a strong sense of responsibility, being innovative, resourceful, open, and responsive to changes. It has created in me an interest in lifelong learning.

I now understand technical terms related to manufacturing, using various mechanical instruments, basic working of lathe and boring machines. I also got an insight about the internal working of the industry.

This internship provided me with a professional learning experience that offers meaningful, practical work related to manufacturing.